

Project Summary

Students will select one real location around Universiti Teknologi Malaysia (UTM), capture their own reference photos, and develop an interactive 3D campus tour based on the selected environment. The application must be developed using Three.js and run in a web browser. Blender may be used to create, modify, or enhance 3D models.

Submission mode: Live demo link, source code repository, report, and presentation

1. Project Overview

This mini project requires each group to design and develop a browser-based interactive campus tour using Three.js. The project should combine real-world observation, multimedia content, and interaction design.

- Students must choose one physical location around UTM, such as a faculty building, library, landmark, laboratory area, food court, bus stop, garden, lake area, or student activity space.
- Students must visit the location and take their own reference photos. These photos will be used to guide the 3D environment, textures, signage, and layout.
- Students must recreate the selected environment as an interactive 3D scene using Three.js. Blender is allowed for modelling, but the final application must run through a web browser.
- The final output should not be a static 3D model only. It must function as a campus tour that allows users to explore, interact, and learn about the selected place.

2. Project Task

In groups of 3 to 4 students, select one real location around UTM. Visit the location, take your own reference photos, study its layout and important visual features, and develop an interactive 3D campus tour based on the selected environment.

- The selected location must be approved by the lecturer to avoid duplication between groups.
- Each group must produce a meaningful campus tour experience, not only a decorative 3D scene.
- The application must be implemented using Three.js and must run in a modern web browser.
- Blender may be used to build or edit models. Other supporting tools may be used with proper acknowledgement.
- The application must be deployed online using Vercel, or another hosting platform.

3. Minimum Requirements

Component	Minimum Requirement
Location selection	Choose one real UTM main location and a sub location extended from the main. The location must be photographed and documented by the group.
Photo evidence	Submit at least 5 original reference photos taken by the group. The photos should show the environment, important objects, signage, and layout.
3D environment	Recreate the selected environment in Three.js with recognizable spatial arrangement and visual similarity to the real place.
3D objects	Include at least 5 meaningful objects, such as building elements, signboards, trees, benches, roads, lamp posts, doors, banners, vehicles, or other location-specific items.
Navigation	Allow the user to explore the scene using suitable controls, such as mouse and keyboard.

Component	Minimum Requirement
Interaction	Include at least 3 interactive elements. Examples: clickable hotspot, information pop-up, animated door, lighting toggle, direction marker, quiz point, audio trigger, object highlight, or camera movement.
Information points	Provide location information using labels, pop-up panels, narration, images, or short descriptions.
User interface	Include a clean UI with a welcome screen, instruction panel, navigation guide, or menu.
Responsiveness	The application should run correctly in a desktop browser and adapt reasonably to different screen sizes.
Deployment	Submit a live URL and source code repository.

4. Deliverables

No.	Deliverable	Details
1	Live web application	Public URL to the deployed Three.js campus tour.
2	Source code repository	GitHub or equivalent repository containing source code, assets, and README file.
3	Project report	PDF or DOCX report following the required structure.
4	Reference photo evidence	Original photos taken by the group. Include them in the report or a clearly labelled folder.
5	Presentation slides	Short presentation for project demonstration and explanation.
6	Individual contribution form	Each member must clearly state their tasks and contribution percentage.

5. Report Structure

The project report should be concise but complete. Suggested length: 8 to 15 pages, excluding appendices.

Section	Required Content
1. Introduction	Project title, selected UTM location, group members, and short project summary.
2. Location Study	Reason for choosing the location, location background, target users, and purpose of the tour.
3. Photo Documentation	Original photos taken by the group with captions and explanation of how they guided the design.
4. Design Plan	Sketch, layout, storyboard, tour route, information points, UI plan, or interaction plan.
6. Multimedia and Interaction	Explanation of interactive features, media integration, hotspots, labels, audio/video, and user flow.
8. Individual Contribution	Table showing member responsibilities and contribution.
9. Conclusion	Summary of achievement and possible future improvements.

6. Appendix A: Individual Contribution Template

Member Name	Role / Main Tasks	Evidence	Contribution %